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Brain size and ecology in small mammals

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(With 3 figures in the text)

Relative brain size (measured as gross brain size after body size effects are removed) differs systematically between families of rodents, insectivores and lagomorphs. The Sciuridae have the largest relative brain size, the Soricidae and Bathyergidae the smallest.

These differences complicate comparisons between relative brain size across phylogenetically diverse species and attempts to relate differences in relative brain size to ecological variables. To overcome these problems, best fit relationships were estimated *for each family*, and values for each *genus* were expressed as deviations from the lines of best fit. We refer to these values as Comparative Brain Size (CBS).

Differences in CBS are related to differences in habitat type (forest-dwelling genera have larger CBS' than grassland forms), in diet (folivores have smaller CBS' than generalists or insectivores, frugivores and granivores), in zonation (arboreal genera have larger CBS' than terrestrial ones) and in activity timing (nocturnal genera have larger CBS' than dirurnal ones). However, these ecological categories are interrelated and, when the effects of other ecological differences are taken into account using analyses of variance, only the differences associated with diet, and possibly habitat remain.

These results are discussed and compared with previous analyses of relative brain sizes among primates and bats.

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Introduction

Variation among species in relative brain size is commonly regarded as a product of differing levels of behavioural and sensory complexity being favoured by natural selection in species occupying different ecological niches (Simpson, 1949; Rensch, 1959; Jerison, 1973). There have, however, been few attempts to identify the ecological factors that select for small versus large brain size.

From comparisons of encephalisation in a large sample of primates and insectivores, Bauchot & Stephan (1966, 1969) concluded that increased habitat complexity, specialized diets and nocturnality were associated with relatively large brains. Jerison (1973), from his extensive survey of vertebrate brains, also suggested that a shift in the "adaptive zone" of early mammals, from diurnality to nocturnality, was associated with an increase in their relative brain sizes. In the Chiroptera, Pirlot & Stephan (1970) suggested that variation in relative brain size was associated with different dietetic preferences, and this result has recently been amplified by Eisenberg & Wilson (1978) who showed that in this group frugivores have the largest relative brain sizes. There are few studies on other small mammals, though Compoint-Monmignaut (1973) measured encephalisation in 31 species of rodents and remarked on the homogeneity of the sample which she attributed to the homogeneity of their ecology.

In this paper the ecological factors that are associated with variation in relative brain size among rodents, insectivores and lagomorphs are investigated. A parallel paper (Clutton-Brock & Harvey, 1980) examines similar relationships within primates. We are concerned here with measures of gross brain size, which are best viewed as statistics for measuring other, closely related attributes. Jerison (1973) lists neurone number, cortical volume and surface area, and also shows that sizes of parts of the brain, especially the cerebellum, neocortex and diencephalon, are very highly correlated with gross brain size.

It is widely known that brain size is closely related to body size. To examine relationships between brain size and ecology it is consequently necessary first to remove differences in brain size caused by that correlation. The calculation of such measures poses a variety of problems. In the past several methods have been used. In 1891, Snell fitted the allometric function:

$$y = ax^b$$

to the relationship between brain weight (y) and body weight (x). This has the logarithmic transformation:

$$\log(y) = \log(a) + b.\log(x)$$

which plots as a straight line with a as the intercept and b the slope or rate of increase of brain weight with body weight. Subsequently, standard values for a and b were calculated and it became common to regard species departures from this line as representing the species degree of encephalisation beyond a basic "somatic" level (Snell, 1891; Dubois, 1898), or similarly, the proportion of brain matter determined by body weight alone (Jerison, 1973). In their study of primates and insectivores, Bauchot & Stephan (1966) used a different technique, measuring deviations from a regression line through "basal insectivores"—a group consisting of the Soricidae, Tenrecidae and Erinaceidae—which have a cluster of characters regarded as evolutionarily primitive (Bauchot & Stephan, 1964).

A common feature of all these methods is the use of a slope calculated from one set of data which acts as the basis for calculating relative brain size in other groups of animals. However, the same slope does not describe relationships at different taxonomic levels (Gould, 1971, 1975; Lande, 1979; Clutton-Brock & Harvey, In press). Characteristic intraspecific slopes lie between 0·2 and 0·3 (Lapique, 1907; Bauchot & Stephan, 1964; Gould, 1971; Lande, 1979), while the mammalian "mouse to elephant" curve is about 0·67, and intermediate taxonomic levels have intermediate values.

Moreover, the height of the intercept on the y axis and the mean of values on the x axis are positively correlated (Gould, 1966, 1971). This means that the common technique of extrapolating regression lines back to a standard weight value in order to compare groups that differ in body weight distribution is not justified. It is clear that in all data sets so far studied, regression slopes differ between taxonomic levels and elevations differ within them. This means that values for relative brain sizes measured by deviations from allometric relationships will vary with the taxonomic level used to set those relationships.

In addition, regression analysis itself provides a biassed estimate of the slope of the relationship between brain and body weight when the latter is measured with an appreciable amount of statistical error (the slope will be shallower, see Harvey & Mace, in press). Major axis (principal component) or reduced major axis methods are more appropriate because they relax assumptions concerning the distribution of error (Kermack & Haldane, 1950; Jolicoeur & Mosimann, 1968).

A further difficulty, but of a different kind, arises through the nature of the data. Species are unevenly distributed among genera so that while many genera have only one or two species, others have 50 or more (e.g. *Rattus*). Since, partly for phylogenetic reasons, congeneric species tend to be both morphologically and ecologically similar, they cannot be treated as statistically independent points (see Harvey & Mace, in press).

In this paper the interaction of taxonomy and ecology on a new measure of relative brain size which avoids the difficulties discussed above are examined. First, because of the non-independence of congeneric species, generic data points were used in the analysis, calculated as the median of component species values (see Clutton-Brock & Harvey, 1980). The relative brain size for any *family* was then measured as the mean deviation from the *common* regression line (Snedecor & Cochran, 1967; Mace, 1979). This line differs from a line through all genera since the slope is set by all family slopes, weighted by their numbers of genera and range of body weight values (Snedecor & Cochran, 1967), and tends, like the family slopes (see above), to have a lower slope than the overall line. The data were also analysed using a second measure of relative brain size for each family, defined as the mean of weightings on the minor axis after subjecting the data to a principal component analysis.

At the *generic* level Comparative Brain Size (CBS) values were estimated as deviations from their own family regression lines since both the overall and common regression lines are biassed estimators for certain genera (Mace, 1979). A CBS measure using principal component analysis has not been included in this paper because regression techniques give slightly shallower slopes than component analysis, and shallower slopes produce less marked relationships between generic CBS values and ecology for the small mammal data. (All conclusions concerning the relationship between generic CBS and ecology hold true when CBS is measured using weightings on the second component of a major axis for the data discussed here.)

Methods

Brain and body weights

Brain weights were calculated from braincase volumes (see Jerison, 1973) measured on skulls stored at the British Museum (Natural History), London. For each species a sample of between 1 and 10 skulls was filled, through the foramen magnum, with 2 mm glass beads whose volume was then determined. Wherever possible, males and females were equally represented in the sample and their measurements were recorded separately. Juvenile or damaged skulls were excluded, and special care was taken to ensure that skulls were thoroughly cleaned out before measuring.

This method was unsuitable for very small skulls (e.g. those of the Soricidae) where X-ray techniques were used to measure skull volumes. Jerison (1969) first developed the method of numeric or graphic double integration for use on fossil or other skulls where only photographs were available for assessing skull volumes; a full description of the technique is given in Jerison (1969). Because of uncertainty over the comparability of the two measuring techniques, cross checks were made by comparing X-ray estimates of cranial capacity with glass bead measures on the same skulls. A sample of skulls was selected over a wide range of body weights and the maximum deviation between the two measures was 10%, but of 17 skulls examined, 11 had deviations less than 5%. Furthermore, there was no evidence that either method consistently underestimated or overestimated (Mace, 1979).

Body weight and body lengths were extracted from the published literature on adult animals both captured and measured in the wild. Pregnant or obese forms were excluded and mean values were calculated for species that undergo marked annual fluctuations in weight. Subspecific or race variants are quite likely to introduce error into such estimates, particularly as in some cases the brain measures come from quite different populations. However, such errors are unlikely to bias results in a particular direction and the increased variances generated will tend to act against any hypothesis being tested.

The relationship between brain and body weight did not differ between males and females. Mean values were therefore used for the two sexes throughout the analyses. Data were available for 261 species in 117 genera (Table I).

Ecological classifications

Species were allocated to one of four ecological categories according to descriptions of their ecology in published works. The basis for classification is fully described in Mace (1979) and a brief account is given below.

Vegetation

Vegetation type in species' habitats was classified according to the degree of cover and diversity of form. Forest and woodland dwellers were contrasted with grassland forms.

Diet type

Species were classified according to the most common item in their diet when that item formed over 50% of their diet. These categories were then sorted into three others on the basis of the way in which food items were likely to be distributed throughout the foraging range. Thus, insects, fruits and seeds, which are generally clumped in distribution and require specific food searching behaviour, form a distinct category from foliage which is abundant and more evenly distributed. Animals documented as frugivores, insectivores or granivores were combined into one category and folivores into a second. A third category, "generalist", describes species that are largely unselective in their diet, and which consume all plant parts and sometimes insects and carrion as well.

TABLE I

Ecological classifications, body weight, body length and brain weight data for species used in the analysis

1*	2345	6	7	8
RODENTIA				
Aplodontidae				
Aplodontia rufa	—T–	806·0	—	7·63
Sciuridae				
Sciurus aberti	WIAD	626·5	268·0	7·44
Sciurus aureogaster	WIAD	—	251·9	8·40
Sciurus carolinensis	WIAD	574·5	353·2	7·76
Sciurus granatensis	WIAD	400·0	230·0	6·50
Sciurus niger	WIAD	809·9	278·9	9·31
Sciurus vulgaris	WIAD	331·8	214·2	6·34
Tamiasciurus hudsonicus	WIAD	188·9	184·0	4·42
Funambulus pennanti	WIAD	85·9	—	2·69
Ratufa affinis	WIAD	982·5	335·0	10·44
Ratufa bicolor	WIAD	1320·0	352·5	11·60
Protoxerus stangeri	WI—	639·5	263·0	11·33
Funisciurus anerythrus	W—	—	331·6	4·73
Funisciurus carruthersi	WI—	268·0	229·1	5·07
Funisciurus pyrrhops	WI—	200·0	200·0	4·97
Paraxerus cepapi	–GTD	216·0	—	3·44
Heliosciurus gambianus	WIA–	295·0	191·9	4·58
Heliosciurus rufobrachium	WIA–	326·5	227·0	6·34
Heliosciurus ruwenzorii	WIA–	291·0	226·1	5·99
Callosciurus caniceps	WIA–	240·0	205·0	6·61
Callosciurus nigrovittatus	WIA–	202·0	210·0	5·03
Callosciurus notatus	WIA–	209·5	187·9	5·50
Callosciurus prevostii	W–A–	439·0	250·0	7·11
Rhinosciurus laticaudatus	WITD	240·0	207·5	4·87
Dremomys rufigemis	–I—	240·0	205·0	6·00
Tamiops macclellandii	WI—	39·0	98·0	2·54
Xerus erythropus	WIT–	722·5	380·0	7·27
Xerus inaurius	–GT–	400·0	—	8·24
Xerus rutilus	–GT–	317·5	222·5	5·68
Spermophilopsis leptodactylus	–GTD	—	244·0	6·77
Marmota flaviventris	GFTD	5000·0	424·5	10·88
Marmota monax	GFTD	4198·5	—	11·24
Cynomys gunnisoni	GFTD	796·3	277·5	5·75
Cynomys leucurus	GFTD	992·1	207·4	6·28
Cynomys ludovicianus	GFTD	793·5	294·4	6·60

*Key to numbers of columns:

1. Species name.
2. Vegetation type: W, woodland; G, grassland or open.
3. Diet: I, insectivores, frugivores or granivores; F, folivores; G, generalists.
4. Zonation: A, arboreal; T, terrestrial; F, fossorial.
5. Activity: N, nocturnal; D, diurnal.
6. Body weight (gs).
7. Body length (mms).
8. Brain weight (gs).

[*Continued overleaf*

TABLE I—*continued*

1*	2345	6	7	8
Ammospermophilus harrisii	GFTD	138·8	—	2·83
Ammospermophilus leucurus	GITD	105·9	152·5	2·92
Spermophilus beecheyi	GFTD	587·5	—	5·71
Spermophilus beldingi	GFTD	262·9	208·0	3·87
Spermophilus columbianus	GFTD	453·6	327·0	4·09
Spermophilus franklinii	GFTD	455·5	245·9	4·41
Spermophilus lateralis	GFTD	166·1	175·0	3·60
Spermophilus nelsonii	GFTD	70·0	—	2·90
Spermophilus parryi	GFTD	744·6	279·5	5·73
Spermophilus richardsonii	GFTD	316·0	—	3·76
Spermophilus tereticaudus	GFTD	156·6	—	2·41
Spermophilus townsendii	GFTD	199·0	177·0	2·65
Spermophilus tridecemlineatus	GFTD	175·0	—	3·02
Spermophilus undulatus	GFTD	725·3	—	5·43
Tamias striatus	W—D	94·3	161·7	2·76
Eutamias amoenus	WI–D	50·8	113·5	1·98
Eutamias dorsalis	W—D	62·2	123·7	2·57
Eutamias minimus	WI–D	45·3	121·1	2·19
Eutamias palmeri	W—D	60·8	—	2·58
Eutamias panamintinus	W—D	51·2	—	2·42
Eutamias quadrimaculatus	WI–D	82·3	143·5	2·75
Eutamias quadrivittatus	—–D	62·3	128·0	2·29
Eutamias speciosus	WI–D	59·9	123·0	2·45
Eutamias townsendii	WI–D	89·4	144·5	3·03
Petaurista elegans	WIAN	924·0	188·5	8·35
Petaurista petaurista	WIAN	2023·0	374·7	12·32
Glaucomys sabrinus	WIAN	174·0	153·5	3·60
Glaucomys volans	WIAN	73·1	—	2·39
Hylopetes spadiceus	WIAD	—	150·1	2·46
Pteromyscus pulverulentus	W–AD	—	227·7	4·14
Iomys horsfieldii	W–AD	155·5	185·5	3·97
Aeromys tephromelas	W–AD	1189·0	380·0	10·93
Geomyidae				
Geomys breviceps	—F–	138·8	156·8	2·29
Geomys bursarius	–FF–	202·1	177·3	2·63
Thomomys bottae	—F–	140·7	156·0	2·00
Thomomys talpoides	—F–	120·7	205·0	1·98
Heteromyidae				
Perognathus baileyi	–ITN	28·3	93·0	1·32
Perognathus flavus	–ITN	8·8	—	0·89
Perognathus formosus	–ITN	20·9	194·2	1·13
Perognathus hispidus	–ITN	39·0	—	1·27
Perognathus longimembris	–ITN	8·3	135·6	0·93
Perognathus pencillatus	–ITN	16·0	77·5	1·06
Microdipodops megacephalus	–ITN	14·1	160·8	1·19
Dipodomys agilis	–ITN	61·4	—	1·93
Dipodomys deserti	–ITN	130·2	334·0	2·33
Dipodomys heermani	GITN	60·0	112·0	1·96

[*Continued overleaf*

1*	2345	6	7	8
Dipodomys merriami	-ITN	37·8	243·0	1·64
Dipodomys microps	-ITN	59·9	273·7	1·80
Dipodomys ordii	-ITN	54·0	242·5	1·97
Dipodomys spectabilis	GITN	142·5	—	2·29
Liomys pictus	-ITN	46·5	—	1·36
Liomys salvinii	WITN	44·9	—	1·36
Heteromys desmerestianus	WITN	72·2	—	1·67
Cricetidae				
Oryzomys longicaudatus	-ITN	26·6	—	1·26
Oryzomys palustris	-ITN	47·1	—	1·47
Nyctomys sumichrasti	—AN	60·0	—	1·67
Ototylomys phyllotis	WI-N	63·0	—	1·92
Reithrodontomys megalotis	-I-N	10·7	71·4	0·99
Peromyscus boylii	GITN	16·3	—	1·37
Peromyscus californicus	GITN	34·4	—	1·54
Peromyscus crinitus	-ITN	15·4	83·1	1·22
Peromyscus eremicus	-ITN	21·3	93·5	1·23
Peromyscus floridanus	WITN	27·3	98·3	1·54
Peromyscus gossypinus	-ITN	26·6	104·7	1·42
Peromyscus leucopus	WITN	21·8	94·7	1·24
Peromyscus maniculatus	-ITN	19·9	94·7	1·19
Peromyscus megalops	-ITN	71·0	—	1·80
Peromyscus mexicanus	-ITN	53·4	125·3	1·63
Peromyscus nuttalli	WITN	21·5	—	1·28
Peromyscus polionotus	GITN	14·0	—	1·12
Peromyscus thomasi	-ITN	77·0	—	2·01
Peromyscus truei	-ITN	30·0	—	1·36
Holochileus sciureus	GG-N	112·3	—	2·00
Sigmodon hispidus	GGTN	132·5	—	1·73
Neotoma albigula	-GTN	223·1	—	2·76
Neotoma cinerea	-GTN	331·3	212·3	3·24
Neotoma floridana	WGTN	257·7	212·9	3·63
Neotoma fusciper	WGTN	193·3	—	3·25
Neotoma microps	-GTN	378·3	—	3·25
Cricetulus triton	WGT-	25·0	—	1·88
Mystromys albicaudatus	—-N	80·0	—	1·98
Dicrostonyx groenlandicus	GFT-	68·4	—	1·31
Dicrostonyx torquatus	W-T-	73·0	—	1·43
Synaptomys cooperi	GFT-	23·8	104·3	1·18
Myopus schisticolor	W—	33·5	—	1·15
Lemmus lemmus	GFT-	68·9	—	1·45
Lemmus trimucronatus	GFT-	69·6	—	1·49
Clethrionomys gapperi	WGT-	16·9	—	1·15
Clethrionomys glareolus	WGT-	17·9	106·6	1·11
Clethrionomys rufocanus	WGT-	38·3	—	1·20
Clethrionomys rutilus	WGT-	17·6	—	1·15
Arvicola terrestris	-G-D	168·3	194·0	2·23
Ondatra zibethica	-G-N	1372·9	283·7	5·32
Neofiber allenyi	G—N	270·5	202·0	3·31

TABLE I—*continued*

1*	2345	6	7	8
Microtus agrestis	GFTN	22·3	—	1·16
Microtus arvalis	GFTN	30·4	114·0	1·14
Microtus californicus	GFTN	43·3	—	1·37
Microtus guentheri	GFTN	51·3	—	1·28
Microtus longicaudatus	GFTN	47·4	121·6	1·29
Microtus montanus	GFTN	39·7	129·0	1·28
Microtus ochrogaster	GFT–	43·8	—	1·30
Microtus oregoni	GFT–	22·0	—	1·11
Microtus pennsylvanicus	GFTN	37·3	118·7	1·36
Microtus pinetorum	GFT–	24·9	101·6	1·15
Microtus townsendii	GFT–	45·8	—	1·47
Lagurus lagurus	–FTN	20·1	92·8	1·01
Gerbillus campestris	—TN	27·5	90·9	1·49
Gerbillus dasyurus	—TN	—	83·3	1·32
Gerbillus gleadowi	–GTN	26·1	—	1·28
Gerbillus nanus	–GTN	14·0	63·9	1·19
Gerbillus paeba	–GTN	24·0	—	1·43
Gerbillus pyramidum	–GTN	40·9	102·0	1·47
Tatera afra	—TN	65·0	—	2·15
Tatera brantsi	—TN	91·7	—	2·15
Tatera indica	–GTN	139·7	—	2·43
Meriones hurrinae	–GTD	71·3	—	1·56
Meriones unguiculatus	—TD	57·4	—	1·72
Spalacidae				
Spalax ehrenbergi	——N	197·0	—	2·47
Rhizomyidae				
Tachyorycetes splendens	——N	234·0	198·0	2·47
Muridae				
Micromys minutus	W—N	6·4	59·1	0·88
Apodemus flavicollis	WI—	32·3	103·3	1·29
Apodemus sylvaticus	WI–N	18·6	87·6	1·17
Thamnomys cometes	W–AN	—	117·0	1·55
Thamnomys rutilans	W–AN	50·2	111·8	1·55
Mesembriomys gouldii	—AN	1110·0	—	5·18
Arvicanthis niloticus	–GT–	84·5	—	1·92
Lemniscomys striatus	W–T–	46·2	103·0	1·44
Rhabdomys pumilio	GG–D	—	109·0	1·28
Hybomys trivirgatus	WGTD	57·5	120·0	1·69
Aethomys chrysophilus	–GTN	117·0	143·3	1·84
Aethomys hindei	WGTN	146·3	—	2·01
Aethomys namaquensis	——N	79·4	—	1·48
Rattus assimilis	W——	108·5	—	2·34
Rattus exulans	GGT–	60·3	145·5	1·48
Rattus fuscipes	W—N	135·5	157·0	2·22
Rattus muelleri	WGT–	311·6	231·8	3·50
Rattus sabanus	WG—	332·7	229·3	3·98
Rattus lutreolus	–GT–	92·7	151·3	2·23

[*Continued overleaf*

1*	2345	6	7	8
Rattus bowersi	WGT–	360·7	253·4	4·63
Rattus cutchicus	WG—	50·0	—	1·67
Rattus edwardsi	WGAN	332·1	231·4	4·81
Rattus niobe	——	42·1	—	1·69
Rattus rajah	W—	134·1	—	2·51
Rattus ruber	——	101·0	—	2·29
Rattus sordidus	——	—	178·0	2·29
Rattus surifer	W—	159·1	—	2·57
Rattus tunneyi	——	—	156·0	2·00
Rattus verecundus	W—	59·4	—	2·07
Rattus whiteheadi	W—	46·3	—	1·62
Pseudomys australis	—TN	50·0	—	1·75
Pogomys sevia	W—	61·4	—	2·08
Melomys cervinipes	W–AN	70·0	—	1·90
Melomys levipes	W–AN	83·4	—	2·45
Melomys rubex	W–AN	49·7	—	1·86
Melomys rufescens	W–AN	54·7	—	1·88
Malacomys edwardsi	WI–N	60·0	—	1·77
Mus booduga	G—	14·6	—	0·90
Mus musculoides	WITN	—	60·8	0·96
Mus platythrix	G—	29·2	—	1·09
Leggadina heermansburgensis	GITN	—	67·0	1·08
Lophuromys flavopunctatus	WITN	60·0	122·0	1·60
Lophuromys sikapusi	WIT–	60·7	118·3	1·72
Notomys alexis	—N	37·0	—	1·55
Acomys wilsonii	—TN	—	80·0	1·17
Uranomys ruddi	—N	33·6	93·1	1·36
Beamys hindei	W–TN	—	150·0	1·70
Saccostomus campestris	WGTN	—	117·4	1·41
Cricetomys gambianus	WGTN	1150·0	375·4	7·16
Praomys daltoni	W—	35·0	—	1·27
Praomys morio	WGAN	46·5	—	1·43
Praomys tulbergi	WGTN	37·2	68·9	1·45
Otomys angoniensis	GGTD	95·0	—	1·01
Otomys irroratus	–FTD	141·0	167·0	1·97
Pogonomys mollipilosus	W—D	73·9	—	2·02
Pogonomys sylvestris	W—	41·2	—	1·72
Chiropodomys gliroides	W–A–	25·3	—	1·29
Hydromys chrysogaster	–I—	698·0	304·0	4·92
Zapodidae				
Zapus hudsonicus	GI–N	17·9	85·2	1·02
Zapus princeps	–I–N	24·5	94·3	1·09
Napoezapus insignus	WI–N	23·2	88·3	1·07
Dipodidae				
Jaculus jaculus	—TN	55·4	102·4	1·80
Hystricidae				
Atherurus africanus	W—	2685·0	—	19·06

TABLE I—*continued*

1*	2345	6	7	8
Erethizontidae				
Erethizon dorsatum	WGT–	5397·0	—	25·19
Caviidae				
Cavia aperea	GFTD	524·9	—	5·86
Galea musteloides	GFTD	375·0	—	3·31
Dinomyidae				
Dinomys branickii	—N	14000·0	—	27·54
Dasyproctidae				
Dasyprocta agouti	——	3000·0	—	23·37
Dasyprocta pratti	——	900·0	—	11·52
Chinchillidae				
Lagostomus maximus	–G–N	4687·5	—	17·11
Capromyidae				
Geocapromys ingrahami	–G–N	717·3	—	5·05
Myocastor coypus	–G—	—	—	17·68
Octodontidae				
Spalacopus cyanus	——	93·0	—	2·17
Echimyidae				
Proechimys semispinosus	WGTN	450·0	230·0	3·59
Hoplomys gymnurus	——	330·0	—	4·72
Thryonomyidae				
Thryonomys swinderianus	GFT–	1625·0	—	13·87
Bathyergidae				
Cryptomys hottentotus	—F–	98·4	—	1·76
Heliophobius argentocinereus	—F–	160·0	154·9	2·02
Bathyergus suillus	—F–	1175·0	130·1	4·34
Heterocephalus glaber	—F–	60·8	130·1	1·11
Ctenodactylidae				
Pectinator spekei	–G–D	180·0	—	2·52
Massoutiera mzabi	–G–D	182·6	—	2·91
INSECTIVORA				
Erinaceidae				
Erinaceus albiventris	–ITN	280·0	—	2·24
Erinaceus europaeus	–ITN	697·0	—	3·77
Talpidae				
Neurotrichus gibbsii	–IF–	10·0	71·7	0·91
Scalopus aquaticus	–I–N	39·6	—	1·48
Scapanus townsendii	–IF–	—	215·9	2·02
Talpa europaea	–IF–	80·5	—	1·69
Condylura cristata	–IF–	50·0	—	1·37
Chrysochloridae				
Chrysochloris stuhlmanni	–ITN	60·3	127·7	1·37

1*	2345	6	7	8
Soricidae				
Blarina brevicauda	–ITN	18·0	—	0·17
Crocidura bicolor	–ITN	5·6	—	0·13
Crocidura russula	–ITN	9·9	—	0·19
Cryptotis parva	–ITN	5·3	—	0·12
Microsorex hoyi	–ITN	3·5	—	0·08
Neomys fodiens	–I–N	17·5	—	0·29
Notiosorex crawfordii	–ITN	6·0	62·4	0·15
Sorex araneus	–IT–	7·5	—	0·25
Sorex arcticus	–IT–	8·2	73·2	0·27
Sorex cinereus	–IT–	3·9	—	0·17
Sorex fumeus	–IT–	8·9	74·4	0·21
Sorex minutus	–IT–	3·9	—	0·15
Sorex obscurus	–IT–	6·3	—	0·23
Sorex ornatus	–IT–	5·0	—	0·12
Sorex palustris	–I—	11·9	78·9	0·31
Sorex trowbridgii	–IT–	5·3	—	0·21
Sorex vagrans	–IT–	5·3	63·6	0·17
Suncus etruscus	–IT–	2·2	—	0·08
Suncus murinus	–IT–	35·0	138·0	0·39
Surdisorex pollulus	–IT–	18·6	91·4	0·40
LAGOMORPHA				
Ochotonidae				
Ochotona hyperborea	–GTN	120·0	—	2·22
Ochotona princeps	–GTN	169·0	—	2·39
Ochotona rufescens	–GTN	250·0	216·2	2·75
Leporidae				
Lepus americanus	–FTN	1500·0	—	9·72
Lepus arcticus	–FTN	4231·0	680·0	15·80
Lepus californicus	–FTN	2314·9	532·2	13·23
Lepus capensis	–FT–	1895·0	—	11·33
Lepus europaeus	–FT–	1954·0	—	14·35
Lepus nigricollis	–FTN	1961·0	—	11·49
Lepus timidus	–FT–	2716·0	—	13·49
Oryctolagus cuniculus	–FT–	1411·8	—	8·98
Sylvilagus audubonii	–FT–	792·5	—	6·07
Sylvilagus bachmani	–FT–	712·4	—	6·58
Sylvilagus floridanus	–FT–	1036·0	—	7·56

Zonation

This specifies the region in which animals are generally active and in which they forage. Arboreal forms which live and climb in trees were contrasted with terrestrial, ground dwelling species.

Activity

Species were classified as either predominantly nocturnal or diurnal.

Analysis

Generic brain weights, body weights and body lengths were calculated from the median of their component species measures. Body length data were used to test predictions about differing gut sizes (see below). The values were then normalized using logarithmic transformations. Analysis of covariance was then used to test for differences in slope or elevation between families (Snedecor & Cochran, 1967). The common regression line (see above) and common principal axis slope (calculated using the maximum likelihood estimate—see Harvey & Mace, in press) were then established.

Relative brain size for each family was then calculated as (i) the mean generic deviation of it's constituent genera from the common regression line and (ii) the mean weighting of genera on the minor axis orthogonal to the common major axis. These values were compared with median family

TABLE II

Slopes and elevations (or weighting on minor axis) for $\log_e$ *(body weight) and* $\log_e$ *(body length) regressed on* $\log_e$ *(brain size)*

Family	No. of genera	Regression analysis Slope	Regression analysis Elevation	Principal axis analysis Slope	Principal axis analysis Elevation	% variance
(a) *Body weight analyses*						
Sciuridae	23	0·42	−0·73	0·43	−0·80	84
Geomyidae	2	0·76	−3·02	0·76	−3·02	—
Heteromyidae	5	0·29	−0·67	0·29	−0·69	75
Cricetidae	23	0·35	−0·90	0·35	−0·92	86
Muridae	23	0·39	−1·05	0·40	−1·06	95
Zapodidae	2	0·12	−0·31	0·12	−0·31	—
Caviidae	2	1·69	−8·84	1·69	−8·84	—
Echimyidae	2	−0·88	6·65	−0·88	6·65	—
Bathyergidae	4	0·43	−1·52	0·43	−1·53	96
Ctenodactylidae	2	9·63	−49·01	9·63	−49·01	—
Talpidae	4	0·29	−0·74	0·29	−0·75	95
Soricidae	9	0·72	−3·07	0·76	−3·26	86
Leporidae	3	0·76	−3·25	0·78	−3·38	95
(b) *Body length analyses*						
Sciuridae	23	1·29	−5·25	1·49	−6·34	81
Geomyidae	2	−2·64	14·41	−2·64	14·41	—
Heteromyidae	3	0·80	−3·73	0·83	−3·93	89
Cricetidae	11	1·23	−5·44	1·31	−5·81	91
Muridae	19	1·03	−4·43	1·07	−4·61	94
Zapodidae	2	−0·67	3·06	−0·67	3·06	—
Bathyergidae	3	1·66	−7·82	1·73	−8·18	95
Talpidae	2	0·72	−3·17	0·72	−3·17	—
Soricidae	4	1·20	−6·70	1·49	−8·01	73

For body weight and body length analyses respectively:
Common regression slope = 0·40, 1·14; total regression slope = 0·59, 1·54,
common principal axis slope = 0·39, 1·21; total principal axis slope = 0·64, 2·03.

% variance accounted for is the correlation coefficient squared × 100. To calculate the common principal axis slope a maximum likelihood method was used (Harvey & Mace, in press) and genera with sample sizes less than 3 could not be incorporated; consequently the same sample was used to calculate the total principal axis slope.

body weights in order to describe differences between families in relative brain size when body size differences were removed. Family relative brain sizes, calculated from body weight and body length analyses, were then compared.

To compare ecological types, species were allocated to groups within the ecological categories of vegetation type, diet, zonation and activity timing. For each genus, a median value for brain weight, body weight or body length was taken from each of these sources and it's deviation from the family regression line calculated. For families where genera differed in their classification on any one ecological category, the deviations of the different ecological groups were compared by one-way analysis of variance. When two interrelated categories both influenced relative brain size, their respective influences were isolated by two-way analysis of variance.

Results

Overall and family relationships

The total regression of brain on body weight, using all points separately, gave a slope of 0·59 (a value of 0·67 is characteristic of most cross-mammal studies (Jerison, 1973)), while the equivalent principal axis slope was 0·64 (Table II).

Because of the high proportion of variance accounted for by the within family relationships between brain and body weight, the regression and principal axis results were extremely similar (see Table II). Therefore, only the regression analysis is considered further. Most within family regressions, and the common regression, gave lower slopes than the total regression and most families differed from one another in elevation but not in slope (slope $F_{12,78}=2{\cdot}03$, $P=0{\cdot}05$; elevation $F_{12,90}=36{\cdot}85$, $P<0{\cdot}001$). Two families, the Soricidae and Leporidae, had significantly greater slopes which cause significant overall heterogeneity in slope between families. When brain weight was regressed on body lengths (Table II), families differed in elevation only (slope $F_{8,51}=2{\cdot}30$, not significant; elevation $F_{8,59}=38{\cdot}67$, $P<0{\cdot}001$).

The pattern of family differences was fairly consistent whether body weight or body length was used as the independent variable (Table III, Fig. 2). Only the Talpidae and Heteromyidae changed in relation to the other families: the former were only represented by two genera for body length, while the latter have an unusually long body length for their body weights (Mace, 1979).

The Sciuridae (squirrels) have the greatest relative brain sizes, considerably larger than the closely related Heteromyidae and Geomyidae. Some of the hystricomorph families seem likely to approach similar levels (see Fig. 1) but cannot be compared directly because they either contain or are represented by a single genus. The Leporidae (hares) also have relatively large brains and are large in body size, while the Soricidae (shrews) have small relative brain sizes and small body sizes. These two groups contribute greatly to the overall relationship between the median body weight (logged) of a family and it's relative brain size (Fig. 3). The Sciuridae, Heteromyidae and Zapodidae have the highest relative brain sizes for their body sizes, while the Geomyidae and Bathyergidae have the lowest (Fig. 3).

Differences between ecological groups

Comparative Brain Sizes (CBS') differ between all ecological categories (see Table IV).

(a) Forest dwellers have larger CBS' than grassland forms ($F_{1,51}=14{\cdot}94$, $P<0{\cdot}005$).

(b) Folivores have lower CBS' than generalists which have lower CBS' than granivores, frugivores and insectivores ($F_{2,51}=10{\cdot}81$, $P<0{\cdot}005$).

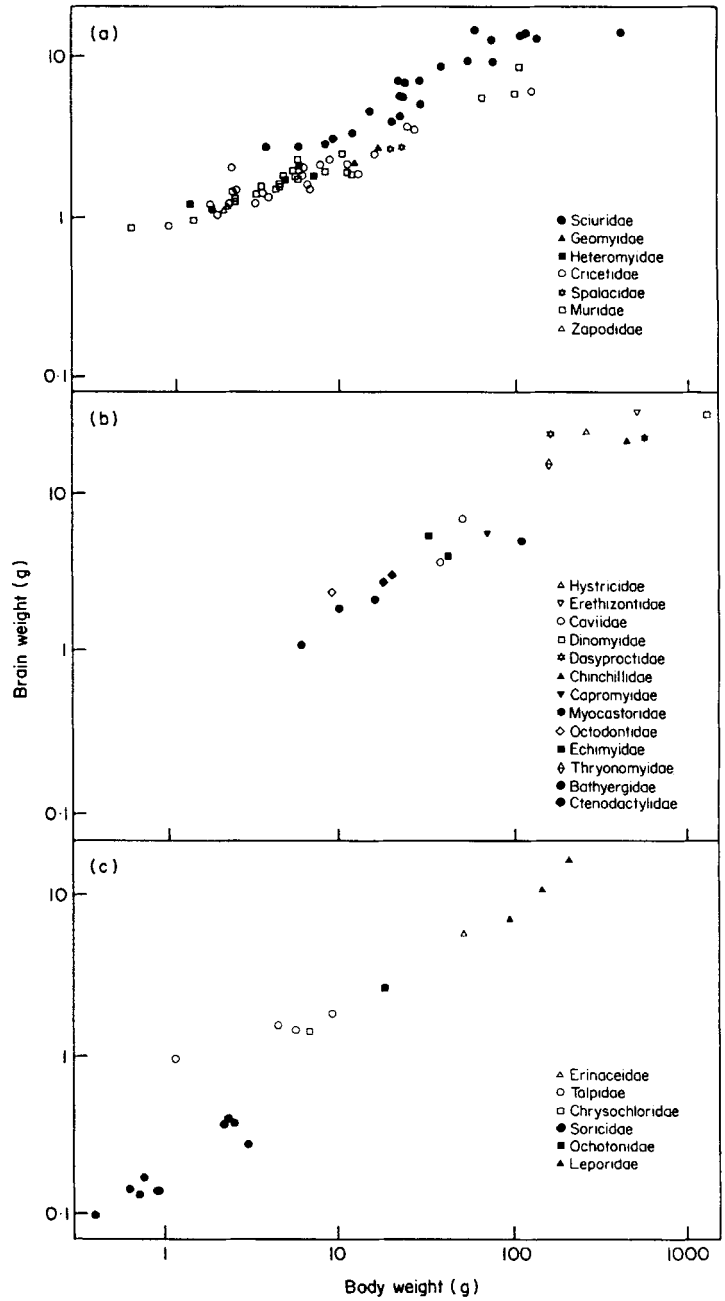


FIG. 1. Brain weight plotted against body weight for generic values (see text) in (a) Sciuromorph and myomorph rodents; (b) Hystricomorph rodents, and (c) Insectivora and Lagomorpha.

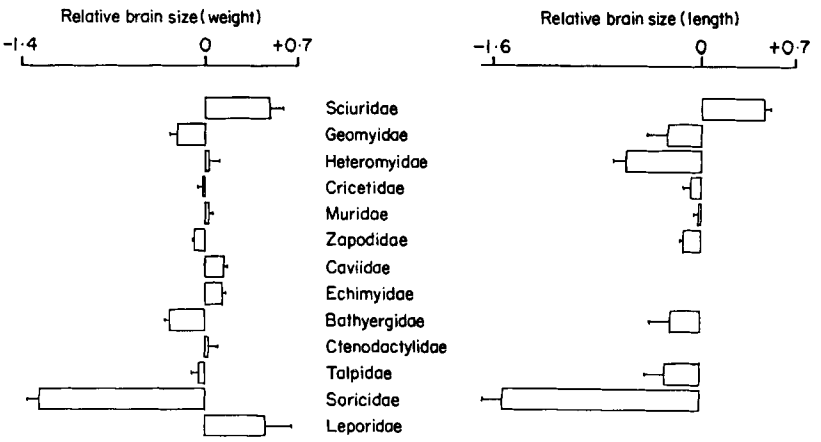


FIG. 2. Relative brain size, measured as mean deviation of genera from the common regression (see text) using both body weight and body length as the independent variables. Error bars denote one standard deviation.

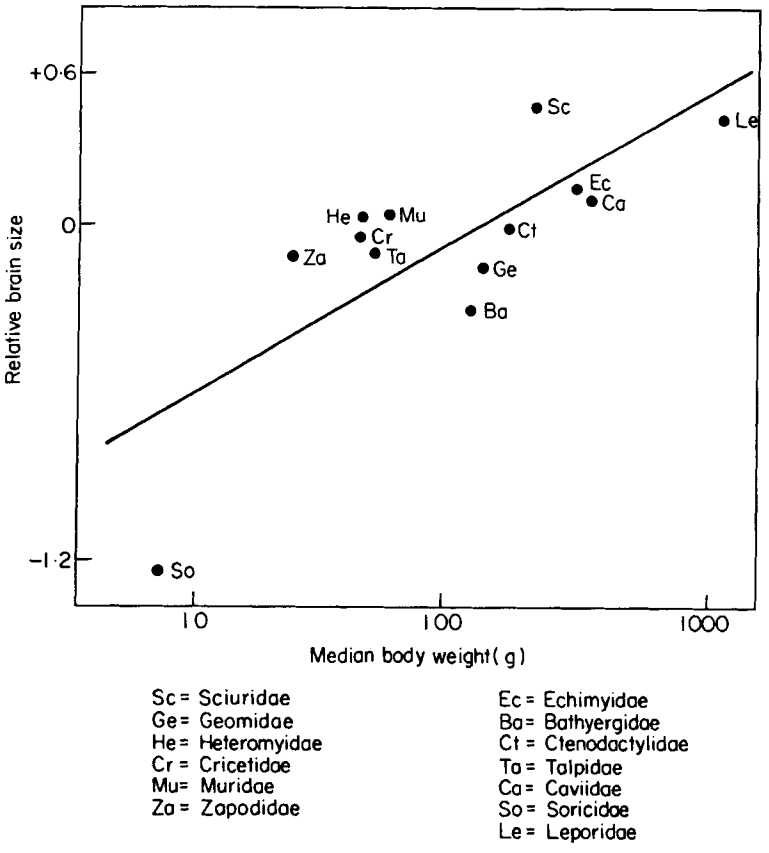


FIG. 3. Relative brain size for each family (see text) regressed on median body weight of component genera.

TABLE III

Relative brain sizes for different families measured as mean generic deviations from the common regression line (or weighting on the minor common axis) of $\log_e$ (brain weight) on $\log_e$ (body weight) or $\log_e$ (body length) (see text for details)

Family	*Body weight* Mean		*Body weight* S.D.		*n*	*Body length* Mean		*Body length* S.D.		*n*
Sciuridae	0·49,	0·47	0·21,	0·19	23	0·47,	0·27	0·23,	0·14	23
Geomyidae	−0·11,	−0·09	0·07,	0·06	2	−0·16,	−0·12	0·20,	0·13	2
Heteromyidae	0·03,	0·02	0·15,	0·13	5	−0·58,	−0·38	0·17,	0·12	3
Cricetidae	−0·02,	−0·01	0·17,	0·15	23	−0·08,	−0·05	0·18,	0·11	11
Muridae	0·03,	0·03	0·12,	0·11	23	−0·02,	−0·01	0·14,	0·10	19
Zapodidae	−0·08,	−0·08	0·02,	0·02	2	−0·13,	−0·09	0·02,	0·01	2
Caviidae	0·15,	0·15	0·31,	0·29	2	—	—	—	—	—
Echimyidae	0·14,	0·14	0·28,	0·26	2	—	—	—	—	—
Bathyergidae	−0·27,	−0·24	0·12,	0·11	4	−0·22,	−0·16	0·26,	0·15	3
Ctenodactylidae	0·02,	−0·14	0·10,	0·10	2	—	—	—	—	—
Talpidae	−0·04,	−0·04	0·11,	0·11	4	−0·27,	−0·17	0·32,	0·24	2
Soricidae	−1·35,	−1·27	0·28,	0·27	9	−1·50,	−0·94	0·25,	0·16	4
Leporidae	0·47,	0·45	0·19,	0·18	3	—	—	—	—	—

n = number of genera in analysis. Mean and standard deviation values are from regression analysis followed by principal axis analysis. S.D. = standard deviation.

TABLE IV

Comparative brain sizes (CBS') within the ecological categories

	Body weight	Body length
Vegetation		
Forest	0·06 ± 0·15 (39)	0·01 ± 0·20 (32)
Grassland	−0·12 ± 0·17 (18)	−0·01 ± 0·20 (12)
	$F_{1,55} = 14{\cdot}94$ $P < 0{\cdot}01$	$F_{1,42} = 2{\cdot}70$ n.s.
Diet		
F.I.G.	0·06 ± 0·13 (26)	0·06 ± 0·20 (23)
Generalist	−0·02 ± 0·18 (19)	0·01 ± 0·18 (14)
Folivore	−0·22 ± 0·12 (9)	−0·20 ± 0·08 (7)
	$F_{2,51} = 10{\cdot}81$ $P < 0{\cdot}01$	$F_{2,41} = 5{\cdot}52$ $P < 0{\cdot}01$
Zonation		
Arboreal	0·07 ± 0·14 (25)	0·04 ± 0·22 (16)
Terrestrial	−0·05 ± 0·18 (32)	−0·05 ± 0·16 (26)
	$F_{1,55} = 6{\cdot}88$ $P < 0{\cdot}01$	$F_{1,40} = 2{\cdot}56$ n.s.
Activity		
Nocturnal	0·02 ± 0·12 (33)	0·02 ± 0·17 (27)
Diurnal	−0·10 ± 0·16 (17)	−0·10 ± 0·15 (14)
	$F_{1,48} = 9{\cdot}35$ $P < 0{\cdot}01$	$F_{1,39} = 4{\cdot}73$ $P < 0{\cdot}05$

For each ecological group, the mean and standard deviation of generic deviations from the family regression line on body length or body weight are shown (together with sample size in parentheses). Differences between groups are tested by one-way analysis of variances. n.s., not significant.

F.I.G., frugivore, insectivore, granivore combined category.

(c) Arboreal forms have larger CBS' than terrestrial forms ($F_{1,55}=6{\cdot}88$, $P<0{\cdot}01$).

(d) Nocturnal groups have larger CBS' than diurnal forms ($F_{1,48}=9{\cdot}35$, $P<0{\cdot}01$).

However, these ecological classifications are not independent of one another. For example, nocturnal species tend to be granivorous, and folivores tend to be both grassland dwellers and terrestrial (Mace, 1979). Two-way analysis of variance was therefore used to isolate the effects of each classification (Table V), and only dietetic differences remain significant. In each case folivores have significantly smaller CBS' than other dietetic groups, although forest and grassland dwellers differ in CBS at the 10% level of significance when dietetic differences are removed. However, dietetic effects seem to be largely responsible for the differences observed within the zonation and activity classifications.

The analysis was repeated using body length in place of body weight and the results are similar (Table IV). In both cases, folivores have relatively small brain sizes compared with other dietetic groups.

TABLE V

Differences in the comparative brain sizes (CBS') of genera classified within two ecological categories. Results of two-way analyses of variance are given

Diets and vegetation	
Diet, vegetation effects removed	$F_{2,41}=4{\cdot}13$ $P<0{\cdot}025$
Vegetation, diet effects removed	$F_{1,41}=3{\cdot}58$ $P<0{\cdot}100$
*Diets and zonation**	
Diet, zonation effects removed	$F_{2,29}=5{\cdot}54$ $P<0{\cdot}001$
	$F_{1,15}=0{\cdot}04$ n.s.
Zonation, diet effects removed	$F_{1,20}=1{\cdot}52$ n.s.
	$F_{1,16}=0{\cdot}29$ n.s.
Diets and activity	
Diet, activity effects removed	$F_{2,34}=5{\cdot}03$ $P<0{\cdot}025$
Activity, diet effects removed	$F_{1,34}=0{\cdot}47$ n.s.

*Because of the lack of arboreal folivores, this analysis could not be done by two-way analysis of variance. One way tests were therefore performed separately within diet and zonation groups.

n.s., not significant.

Discussion

Differences between families

One of the most pervasive features of this analysis was the marked but consistent differences in relative brain size between families. There are two conventional explanations for this phenomenon, and they could both have been operating. First, there has been a progressive increase in relative brain size over evolutionary time associated with morphological divergence from the ancestral stock (Jerison, 1973). Second, differences in relative brain size may be functional and represent adaptations to particular ecological niches. Both are discussed below.

Phylogeny

The detailed phylogeny of the rodents is poorly understood since their radiation in the late Eocene was explosive and gave rise to most of the major lineages, some of which

apparently arose independently more than once (Romer, 1966; Wood, 1955). The difficulty is increased for the hystricomorphs, whose precise taxonomic affinities are still widely disputed (Simpson, 1974; Wood, 1974).

The Insectivores retain many characteristics similar to those of the early mammals and it is among this group that families with very small relative brain sizes, such as the Soricidae, are found. However, other, more specialized, insectivore families such as the Talpidae (moles) and Chrysochloridae (golden moles) have relative brain sizes similar to those of most rodents. Also, the lagomorphs have quite recently diverged and the hares have quite large relative brain sizes.

Among the rodents, the sciuromorphs retain most primitive characteristics, but these have the largest relative brain sizes. Thus, while it may be possible to relate relative brain size variation to phylogenetic divergence at gross levels, differences within finer categories are not easily explained.

Ecology

Most of the major phylogenetic lineages within the sample contain at least one group that has adopted a fossorial way of life. Since this habit must represent one of the most simple sensory and perceptual niches, it is of interest to compare relative brain size in these fossorial families with those of other closely related families.

The Bathyergidae (mole rats) and Geomyidae (gophers) both have small relative brain sizes compared to other families and this becomes most evident when they are compared with other families in their sub-orders, whether or not body size differences are removed (Figs 1 and 3). In both groups, individuals live almost entirely below ground in burrows and so tend to have reduced sensory faculties. Their food consists of roots, bulbs and tubers which are reached by means of feeding burrows (Genelly, 1965; Jarvis & Sale, 1971; Aldous, 1951). Similarly, the Spalacidae and Rhizomyidae (European mole rats) are completely blind although they are known to forage above ground on occasions (Reed, 1958; Nevo, 1961). Both these families have small relative brain sizes compared with other myomorphs (Fig. 1).

Other fossorial families do not have particularly small relative brain sizes, but each of these also shows additional specializations. For example, the cururos (*Spalacopus*; Octodontidae) are social and the groups are nomadic, moving into a new area when the present site becomes depleted of food (Reig, 1970); their brain sizes are quite similar to those of other hystricomorphs. The insectivore moles (Talpidae) have quite large relative brain sizes although the comparison with the primitive shrews may be misleading. However, moles are not deep burrowers and frequently rest at ground level or burrow only through leaf litter (e.g. *Chrysochloris*, Jarvis, 1974; *Scapanus*, Moore, 1933). In addition they are carnivorous, feeding on earthworms and insect larvae which demand more specific food searching behaviour than truly fossorial feeding. To some extent, therefore, relative brain sizes of the fossorial families relate to their habits.

Families whose genera all fall into a single dietetic category can also be compared. When body weight differences are removed (Fig. 3), those families having high relative brain sizes (lying above the regression line)—the Zapodidae, Heteromyidae and Talpidae—are granivores or insectivores, while those with small relative brain sizes are all folivores or generalists. Thus brain size differences due to diet *within* families are also evident in comparisons *between* families.

A third feature which can be seen at the family level is an association of relative brain size with the degree of development of social behaviour. Social living is most highly developed in the Sciuridae and in some of the hystricomorph families, which are also the groups with the largest relative brain sizes (Fig. 1). In Sciurids, for example, complex patterns of social behaviour are present in *Cynomys* (King, 1955), *Citellus* (Dunford, 1977) and *Marmota* (Barash, 1973), and this is paralleled, at least by colonial living, in the South American viczachas (Chinchillidae—Pearson, 1948). Little is known of many other hystricomorph families; the mara (*Dolochotis patagonum*) is reported to be strictly monogamous (Genest & Dubost, 1974) although many caviomorphs seem to be largely asocial in the wild. Most other small rodents have rather unspecialized social behaviour and exhibit the open systems characteristic of stem mammals (Eisenberg, 1966, 1967). Clearly the association of social living with relative brain size is weak, but this may be a consequence of both being related to food resources. However, it would seem to merit further study in a group of mammals whose social systems are better documented. A positive correlation between CBS and social structure is found in primates (Clutton-Brock & Harvey, 1980) where a small sample of monogamous genera have relatively small brains in comparison with those genera living in larger groups than a bonded pair.

Differences within families

Within families, folivores had smaller CBS' than granivores and insectivores. The effect persists between families (see above), and within each of the families whose genera differ in diet when they are analysed separately. Why is this so?

One immediate speculation must be that folivorous species have a larger gut relative to their body weight in order to facilitate breakdown of food with a high fibre content (Vorontsov, 1962; Hladik, 1967). This could increase body weight, irrespective of brain weight, and result in small CBS'. However, when the analysis was repeated using body lengths in place of body weights, folivores still had significantly smaller brain sizes (Table IV). This would tend to refute the hypothesis since there is no reason to suppose that a larger gut (volume increase) would require equivalent skeletal enlargement. It is possible, however, that a larger gut needs more support and that the "shapes" of animals are more constrained than has been allowed for in this discussion.

A relationship between relative brain size and diet has been demonstrated in other mammalian orders. Among primates, a similar association between folivory and small CBS' is found between families and within them using both body weight and body length estimates of CBS' (Clutton-Brock & Harvey, 1980). In the Chiroptera, the largest relative brain sizes are found in frugivorous species and the lowest in aerial insectivores, which hunt by echolocation, employing few visual or olfactory cues (Eisenberg & Wilson, 1978). In both the primates and Chiroptera, frugivores seek energy rich pockets of food which are distributed unevenly in both space and time. The authors of both studies suggest that more complex foraging strategies select for increased brain size. The hypothesis is supported by the data presented in this paper: granivores and insectivores which have relatively large brain sizes must search for their food which is likely to be distributed more unevenly over a larger home range than that of folivores (see refs. in Mace, 1979). In addition, granivores and folivores must learn to be selective in the items they eat. Since greater search and handling times are necessary than for folivores, an ability to distinguish between items is more important. Also, a number of complex behavioural patterns are

associated with granivory. It is common for seeds to be stored in preparation for adverse periods. For example, in the North American flying squirrel (*Glaucomys volans*) nuts are stored in response to changes in daylight length (Muul, 1969). Complex patterns of food caching behaviour have also been described (e.g. *Xerus erythropus*, Ewer, 1965). All these features suggest that granivores, frugivores and insectivores occupy a more complex sensory and perceptual niche in which an increase in sensory modalities afforded by an increased brain size would be advantageous.

There was a weak association between large CBS' and woodland habitats. Forests and woodlands are physically more diverse than grassland and are negotiated by resident species moving in three dimensions. Similar arguments based on niche complexity could provide an explanation for this association.

Overall relationships

An overall exponent of 0·67 relating brain weight to body weight has led many authors to suppose that some kind of surface area effect is involved in the scaling of brain on body weight, although the nature of this association is not immediately apparent (see Gould, 1975). If, however, an exponent of 0·67 is regarded as a stable overall relationship, then the lower exponents within taxa means that the larger members of any taxon must have rather small amounts of brain for their body masses. Passingham (1975, 1978) has discussed this problem and shows that the relationship of brain size to the size of the medulla, and of the brainstem, does not differ between taxonomic levels. Apart from the cranial nerves, most information entering the brain passes through these two regions and therefore the ratio of input and output to brain matter does not differ between different sized forms. Passingham (1975) also suggests that relative brain size should be measured from regressions on these structures rather than on body weight. Radinsky (1967) has suggested using the foramen magnum but this structure has also been used to estimate brain size (Jerison, 1973; Gould, 1975). While these methods could facilitate cross-species comparisons, they do not offer any explanation for the variation in exponents.

Addressing this same problem, Gould (1971) has produced a developmental model whereby individuals increase in body size, but with a lower exponent for brain weight than would be found across taxa, until ultimately a size is reached where the relative brain size is rather small. Such a form would then be particularly susceptible to some developmental anomaly which produces a sudden increase in brain size. This process assumes first, that body size tends to increase, and second, that under selection for increased body size increases in brain size tend to be small (see Lande's (1979) analysis of possible pleiotropic effects of genes influencing both brain and body size for a recent view of this problem). The first of the assumptions is well documented for many taxa (Simpson, 1953) and the advantages of large body size have been reviewed (Bourlière, 1975; Clutton-Brock & Harvey, 1977). As for the second point, there are, causally, several ways in which body size need not be accompanied by similar increases in brain size. For example, cell size increases or a thicker layer of fat do not obviously necessitate improved muscular or sensory control. In addition, it could well be adaptive to have brain tissue that is relatively inert to change since the brain controls much somatic form and function, and a small change in brain tissue could have much more profound effects on an animal than a small change in somatic tissue.

Alternatively, from an ecological viewpoint, the invasion of a new niche is more likely to increase levels of behavioural complexity. This is partly because simpler niches are likely to be already occupied and partly because once a behaviour or sensory pathway has developed it is rarely lost and modification tends to result in greater complexity. In this way, relative brain size or CBS should increase with body size and especially large brains should be characteristic of species that have become secondarily small. Bauchot & Stephan (1969) suggested that the high encephalisation quotient of the Talapoin monkey (*Miopithecus talapoin*) could have arisen in this way.

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